Example 1 - Problem Statement:

Given a system of two particles:

- **Particle A_1:**
  - Mass (m_1): $1\ \mathrm{kg}$
  - Velocity ($\vec{V}_1$): $2\vec{i}\ (\mathrm{m/s})$
- **Particle A_2:**
  - Mass (m_2): $500\ \mathrm{g}$
  - Velocity ($\vec{V}_2$): $-4\vec{i}\ (\mathrm{m/s})$

Determine $\vec{V}_G$ (the velocity of the center of mass) and conclude.

Step-by-Step Solution:

Following the principles from your "Chapter 2: Linear Momentum" notes.

1. Standardize Units (SI Units): Before any calculation, we must ensure all units are in the standard SI system (kilograms, meters, seconds).

- $m_1 = 1\ \mathrm{kg}$
- $m_2 = 500\ \mathrm{g} = 0.5\ \mathrm{kg}$
- $\vec{V}_1 = 2\vec{i}\ \mathrm{m/s}$
- $\vec{V}_2 = -4\vec{i}\ \mathrm{m/s}$

2. Calculate the Total Linear Momentum of the System ($\vec{P}_S$): As per **Section B** of your notes, the total linear momentum of the system is the vector sum of the linear momenta of its particles.

$$\vec{P}_S = \vec{P}_1 + \vec{P}_2$$

$$\vec{P}_S = m_1\vec{V}_1 + m_2\vec{V}_2$$

Now, we substitute the values:

$$\vec{P}_S = (1\ \mathrm{kg}) \cdot (2\vec{i}\ \mathrm{m/s}) + (0.5\ \mathrm{kg}) \cdot (-4\vec{i}\ \mathrm{m/s})$$

$$\vec{P}_S = (2\vec{i}\ \mathrm{kg} \cdot \mathrm{m/s}) - (2\vec{i}\ \mathrm{kg} \cdot \mathrm{m/s})$$

$$\vec{P}_S = \vec{0}\ \mathrm{kg} \cdot \mathrm{m/s}$$

3. Calculate the Total Mass of the System (M_S): The total mass (M_S) is the scalar sum of the individual masses.

$$M_S = m_1 + m_2$$

$$M_S = 1\ \mathrm{kg} + 0.5\ \mathrm{kg}$$

$$M_S = 1.5\ \mathrm{kg}$$

4. Determine the Velocity of the Center of Mass ($\vec{V}_G$): From **Section C** of your notes, the total linear momentum of the system is also defined by the total mass and the velocity of the center of mass:

$$\vec{P}_S = M_S \cdot \vec{V}_G$$

We can rearrange this formula to solve for $\vec{V}_G$:

$$\vec{V}_G = \frac{\vec{P}_S}{M_S}$$

Substituting our calculated values:

$$\vec{V}_G = \frac{\vec{0}\ \mathrm{kg} \cdot \mathrm{m/s}}{1.5\ \mathrm{kg}}$$

$$\vec{V}_G = \vec{0}\ \mathrm{m/s}$$

Conclusion:

The velocity of the center of mass of the system ($\vec{V}_G$) is $\vec{0}\ \mathrm{m/s}$.

This result signifies that the center of mass of the system is stationary. Although the individual particles A_1 and A_2 are moving, their momenta are equal in magnitude ($P_1 = 2\ \mathrm{kg} \cdot \mathrm{m/s}$ and $P_2 = 2\ \mathrm{kg} \cdot \mathrm{m/s}$) and in opposite directions. As a result, the total momentum of the system is zero, and the center of mass does not move.